Orestis Melkonian

Personal Data

Place | Date of Birth:	Athens, Greece | December 1992
email:	melkon.or@gmail.com
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Education

Sep 2019 - Sep 2023	PhD in Computer Science @ **University of Edinburgh**, Scotland *Institute*: Laboratory for Foundations of Computer Science (LFCS) *Supervisor*: Philip Wadler *Co-supervisors*: Brian Campbell, Aggelos Kiayias *Thesis*: Verified Compilation from BitML to Bitcoin: An Agda Odyssey
Sep 2017 - Jul 2019	M.Sc. in Computing Science @ **Utrecht University**, The Netherlands *GPA*: 8.99/10 (*cum laude*) *Specialization*: Programming Technology *Supervisors*: Wouter Swierstra (UU), Manuel M.T. Chakravarty (IOHK) *Thesis*: Formalizing Extended UTxO and BitML Calculus in Agda
Oct 2010 - Apr 2016	B.Sc. in Computer Science @ **National and Kapodistrian University of Athens**, Greece *GPA*: 8.6/10 (*distinction*) *Specializations*: Knowledge and Data Management, Software *Supervisors*: Panos Rondogiannis, Angelos Charalambidis *Thesis*: RHEA - A Reactive, Heterogeneous, Extensible and Abstract Framework for Dataflow Programming
Feb 2014 - Jul 2014	Erasmus exchange student in Computer Science **Universita della Svizzera Italiana**, Lugano, Switzerland *Coordinators*: Ioannis Emiris, Marc Langheinrich

Work Experience

May 2026 - now	Formal Methods Scientist @ **Kodamai**, Scotland *Typed Agentic AI via containers / polynomial functors* *Supervisors*: Neil Ghani, Toby Smithe
Sep 2023 - Apr 2026	Formal Methods Engineer @ **Input Output (IOG/IOHK)** *Formalizing the Cardano ledger & BFT consensus protocols in Agda; developing Agda compilers* *Supervisor*: James Chapman
Feb 2021 - Mar 2023	Haskell Reviewer (part-time) @ **MigaMake**, Singapore *Reviewing code, technical reports, and scientific articles* *Supervisor*: Michal Gajda
Jan 2019 - Jul 2019	Haskell Developer (part-time) @ **QBayLogic**, The Netherlands *Implemented a GUI to inspect optimization steps in the Cλash compiler* *Supervisor*: Christiaan Baaij
Dec 2017 - Dec 2018	Haskell Developer (part-time) @ **Utrecht University**, The Netherlands FormalZ: A tower-defence game for learning Hoare-style specifications *Developed the SMT-based checker between teacher and student formulas* *Supervisor*: Wishnu Prasetya
Jun 2017 - Aug 2017	Research Intern @ **Max Planck Institute for Software Systems (MPI-SWS)**, Germany Software Analysis & Verification Group *Proof mechanization of a novel C/C++11 memory model in Coq* *Supervisor*: Viktor Vafeiadis
May 2016 - May 2017	Technical Intern (*Backend Web Developer*) @ **CERN**, Switzerland Invenio project: *Video workflows for the Cern Document Server (CDS)* *Supervisor*: Ludmila Marian
Oct 2015 - Apr 2016	Research Intern @ **National Center for Scientific Research "Demokritos"**, Athens, Greece Software & Knowledge Engineering Laboratory (SKEL) *Design and implementation of a dataflow framework for robotics* *Supervisor*: Angelos Charalambidis

Research Interests

- Programming Languages, Semantics
- Functional Programming, Compilers
- Formal Verification, Theorem Proving
- (Dependent) Type Theory, Category Theory
- Consensus, Blockchain, Smart Contracts
- Safe AI, Agentic Workflows

Selected Publications

TYPES '26 — **Peregrine: a middle-end for code generation from proof assistants**
E.H.Nielsen, S.Dima, L.Escot, O.Melkonian, H.S-Cholet, J.Chapman, Y.Forster, M.Sozeau, B.Spitters
32^{nd} International Conference on Types for Proofs and Programs

FMBC '25 (ETAPS) — **A readable and computable formalization of the Streamlet consensus protocol**
Mauro Jaskelioff, Orestis Melkonian, James Chapman
6^{th} International Workshop on Formal Methods for Blockchains

FMBC '25 (ETAPS) — **Program logics for ledgers**
Orestis Melkonian, Wouter Swierstra, James Chapman
6^{th} International Workshop on Formal Methods for Blockchains

FMBC '25 (ETAPS) — **Validity, Liquidity, and Fidelity: Formal Verification for Smart Contracts in Cardano**
Tudor Ferariu, Philip Wadler, Orestis Melkonian
6^{th} International Workshop on Formal Methods for Blockchains

NeurIPS '24 — **Learning Structure-Aware Representations of Dependent Types**
Konstantinos Kogkalidis, Orestis Melkonian, Jean-Philippe Bernardy
38^{th} Annual Conference on Neural Information Processing Systems

FMBC '24 (ETAPS) — **Formal specification of the Cardano blockchain ledger, mechanized in Agda**
A.Knispel, O.Melkonian, J.Chapman, A.Hill, J.Jääger, W.DeMeo, U.Norell
5^{th} International Workshop on Formal Methods for Blockchains

FMBC '24 (ETAPS) — **Structured Contracts in the EUTxO Ledger Model**
P.Vinogradova, O.Melkonian, P.Wadler, M.Chakravarty, J.Krijnen, M.Jones, J.Chapman, T.Ferariu
5^{th} International Workshop on Formal Methods for Blockchains

TYPES '23 — **Nominal techniques as an Agda library (Extended Abstract)**
Murdoch J. Gabbay, Orestis Melkonian
29^{th} International Conference on Types for Proofs and Programs

Haskell '22 (ICFP) — **Reasonable Agda is correct Haskell: writing verified Haskell using** AGDA2HS
Jesper Cockx, Orestis Melkonian, Lucas Escot, James Chapman, Ulf Norell
15^{th} International Haskell Symposium. ACM

ISoLA '20 — **Native Custom Tokens in the Extended UTXO Model**
M.Chakravarty, J.Chapman, K.MacKenzie, O.Melkonian, J.Müller, M.Jones, P.Vinogradova, P.Wadler
9^{th} International Symposium On Leveraging Applications of Formal Methods, Verification and Validation

ECOOP '20 — **Reconciling Event Structures with Modern Multiprocessors**
E.Moiseenko, A.Podkopaev, O.Lahav, O.Melkonian, V.Vafeiadis
34^{th} European Conference on Object-Oriented Programming

WTSC '20 (FC) — **The Extended UTXO Model**
M.Chakravarty, J.Chapman, K.MacKenzie, O.Melkonian, M.Jones, P.Wadler
4^{th} Workshop on Trusted Smart Contracts

FARM '19 (ICFP) — **Music as Language: Putting Probabilistic Temporal Graph Grammars to Good Use**
Orestis Melkonian
7^{th} ACM SIGPLAN International Workshop on Functional Art, Music, Modeling, and Design. ACM

FARM '19 (ICFP) — **What Constitutes a Musical Pattern?**
Orestis Melkonian, Iris Yuping Ren, Wouter Swierstra, Anja Volk
7^{th} ACM SIGPLAN International Workshop on Functional Art, Music, Modeling, and Design. ACM

ICSE' 19 — **Having Fun in Learning Formal Specifications**
Wishnu Prasetya, Craig Leek, Orestis Melkonian *et al.*
41^{th} International Conference on Software Engineering: Software Engineering Education and Training. ACM

REBLS' 18 (SPLASH) — **RHEA: A Reactive, Heterogeneous, Extensible and Abstract Framework for Dataflow Programming**
Orestis Melkonian, Angelos Charalambidis
5^{th} ACM SIGPLAN International Workshop on Reactive and Event-Based Languages and Systems. ACM

ESSLLI '18 — **D^3 as a 2-MCFL**
Konstantinos Kogkalidis, Orestis Melkonian
Student Session of the 30^{th} European Summer School in Logic, Language and Information. Springer

Teaching Experience

Sep 2020 - Mar 2021	**Reasoning & Agents** @ University of Edinburgh *Demonstrator & Marker*
Sep 2019 - Mar 2020	**Types and Semantics for Programming Languages** @ University of Edinburgh *Teaching Assistant*
Sep 2019 - Mar 2021	**Introduction to Computation: Functional Programming** @ University of Edinburgh *Teaching Assistant & Tutor & Marker*

Academic Service

Apr 2026	Program Committee @ **FMBC**, International Workshop on Formal Methods for Blockchains, 2026
Apr 2025	Invited Reviewer @ **JFP**, Journal of Functional Programming, 2025
Oct 2025	External Reviewer @ **ITP**, International Conference on Interactive Theorem Proving, 2025
May 2025	Program Committee @ **FMBC**, International Workshop on Formal Methods for Blockchains, 2025
Nov 2024	Artifact Evaluation Committee @ **POPL**: Symposium on Principles of Programming Languages, 2025
Feb 2024	Artifact Evaluation Committee @ **OOPSLA**, 2024
Nov 2023	Artifact Evaluation Committee @ **POPL**: Symposium on Principles of Programming Languages, 2024
Jun 2023	Artifact Evaluation Committee @ **ICFP**: International Conference on Functional Programming, 2023
Feb 2023	Artifact Evaluation Committee @ **ESOP**: European Symposium on Programming, 2023
Nov 2022	Organizer @ **31st Agda Implementors' Meeting (AIM XXXI)**, Edinburgh, Scotland
Oct 2022	Artifact Evaluation Committee @ **POPL**: Symposium on Principles of Programming Languages, 2023
Jun 2022	Artifact Evaluation Committee @ **ICFP**: International Conference on Functional Programming, 2022
Mar 2022	Invited Reviewer @ **TYPES** post-proceedings, 2022
Feb 2022	Artifact Evaluation Committee @ **ESOP**: European Symposium on Programming, 2022
Jul 2021	Invited Reviewer @ **APLAS**: Asian Symposium on Programming Languages and Systems, 2021
Aug 2021	Invited Reviewer @ **MSCS**: Mathematical Structures in Computer Science *special edition in memory of Martin Hofmann*

Awards

Jun 2023	**EuroProofNet** Short-term Scientific Mission @ **Chalmers University of Technology**, Gothenburg, SE *Host: Jean-Philippe Bernardy*
May 2023	**SICSA** Research Scholar Grant @ **36th Agda Implementors' Meeting (AIM XXXVI)**, Delft, NL
Jun 2022	Travel Grant @ **Programming Language Mentoring Workshop (PLMW)**, PLDI, San Diego, US
Jun 2019	Travel Grant @ **ZuriHac: A 3-day Haskell Hackathon**, Zurich, CH
Apr 2019	Travel Grant @ **School and Workshop on Univalent Mathematics**, Birmingham, UK
Aug 2018	Registration Grant @ **European Summer School on Logic, Language & Information (ESSLLI)**, Sofia, BG
Jul 2018	Travel Grant @ **DeepSpec Summer School**, Princeton, US
Jul 2017	Travel Grant @ **CAV Conference**, Heidelberg, DE
Jan 2016	Travel Grant @ **Programming Language Mentoring Workshop (PLMW)**, POPL, Florida, US
Mar 2016	Travel Grant @ **Meddays '16**, Sophia-Antipolis, FR

Languages

Greek:	Native speaker
English:	**Fluent**, *Certificate of Proficiency, University of Michigan*
German:	**Intermediate**, *Zertifikat B1, Goethe Institute*
French:	**Beginner**, *A2.1, Supercomm Suisse SA*

Extra-Curricular Activities

Mar 2014	Electric jazz guitar (Intermediate) **RGT Grade 6 (***distinction***)**, *University of West London - Certificate in Music Performance* *Examiner:* Adam Moore BA(Hons), MMus, PGCE, FLCM

References

Philip Wadler
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